

MONIL ARORA

Vancouver, BC

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🌐 [LinkedIn](#)

📁 [Portfolio](#)

Education

University of British Columbia, Vancouver

Expected 2028

Bachelor of Applied Science in Electrical Engineering

Relevant Coursework: Computing Systems I, Digital Systems Design, Introduction to VLSI Systems, Machine Learning with Engineering Applications, Signals and Systems, Systems and Control (in progress)

Work Experience

Laser Zentrum Hannover e.V. (LZH)

Jan 2026 – Jul 2026

Research Intern | C/C++, STM32, FreeRTOS, LittleFS, YAMCS, Java, Python, HIL

Hannover, Germany

- Developed real-time **C/C++** firmware on **STM32** using **FreeRTOS** to handle laser control, telemetry, and telecommands for the **Moonrise mission**, a lunar 3D-printing project that melts regolith with a laser.
- Upgraded persistent storage to the lightweight **LittleFS** filesystem, providing organized data management and **dynamic wear leveling** to extend flash lifetime.
- Eliminated manual flash-sector allocation by letting **LittleFS** manage data placement while keeping the system clock and boot count intact across resets so the ground station can track every reboot.
- Resolved a build-breaking **784-byte** RAM overflow on a **20 KB** chip by reducing the laser task's **memory footprint** by **60%** and heap allocation by **10%**.
- Enabled reliable **image downlink** by fixing a **watchdog timing conflict** that reset the payload mid-transfer.
- Built **YAMCS** mission control dashboards in **Java** and **Python** for real-time **health monitoring**, data visualization, and command operations during tests.
- Integrated a command-triggered **hardware-in-the-loop (HIL)** camera that captured images and uploaded them to the **YAMCS** mission database with associated metadata.

Sharang Shakti Pvt Ltd

May 2024 – Jun 2024

Embedded Signal Processing Intern | AD9910 DDS, Python, PLL

Gurugram, India

- Brought up an **AD9910 DDS** evaluation board for the **digitization of a legacy analog radar system**, generating high-frequency transmit signals.
- Wrote **Python** to build **14-bit** sample tables for custom waveform playback from on-chip RAM, and stabilized the reference clock (**PLL**) for higher resolution and lower **phase noise**.

Design Team

UBC Orbit | ALEASAT CubeSat

Sep 2026 – Present

AOCS Hardware & Test Engineering Member | HIL, AOCS

Vancouver, BC

- Developing a simulation-to-processor bridge for HIL testing of the attitude and orbit control system (AOCS).

Projects

Self-Balancing Robot | C++, MATLAB, Arduino, ESP32, IMU, PID, Embedded Systems

Jan 2025 – Apr 2025

- Prototyped a two-wheeled **self-balancing robot** on **Arduino**, estimating tilt by fusing gyroscope and accelerometer data with a **complementary filter** and stabilizing the robot with a **PID control loop**.
- Modeled the robot's tilt dynamics and tuned **PID** gains in **MATLAB/Simulink** before hardware integration.
- Paralleled two motor drivers to raise current, cutting start-up duty cycle from **15–20%** to **10%** for finer balance control.
- Built a **Bluetooth** remote to drive the robot and display live **telemetry** on an **LCD**.

Deep Neural Network Accelerator | Verilog, C, Nios II, FPGA

Nov 2024 – Dec 2024

- Implemented a **deep neural network** on a **Nios II** soft-core CPU on a FPGA to classify handwritten **MNIST** digits.
- Designed a **SystemVerilog** dot-product accelerator on the **Avalon** bus for **1.79M multiply-accumulates** per image.
- Streamed **7.2 MB** of network weights from off-chip **SDRAM** to overcome the FPGA's **0.5 MB** on-chip memory limit, with custom **clocking** for reliable SDRAM access.

Reflow Oven Controller | Assembly, 8051

Jan 2024 – Feb 2024

- Programmed an **8051-based reflow oven controller** in **assembly**, sequencing a six-state soldering profile with temperature- and time-triggered transitions.
- Implemented **fault handling** with a start-up abort that cuts heater power if the oven is below **50°C** after **60 s** and a stop button active in every state.
- Built a **Python** interface that plotted live oven temperature streamed over **UART**.

Technical Skills

Languages: C/C++, Assembly, Python, Verilog/SystemVerilog, Java, Shell

Embedded: FreeRTOS, LittleFS, UART, I2C, RS-422, PlatformIO

Microcontrollers & FPGA: STM32, ESP32, Arduino, Nuvoton 8051, Nios II, DE1-SoC FPGA

Tools: Git, Linux, Quartus, MATLAB, YAMCS

Lab & Test: Oscilloscope, multimeter, function generator, soldering, hardware-in-the-loop testing